MA1522 Linear Algebra for Computing
L2 Lesson 22



Tan Kai Meng

Semester 1, 2026/27

Question 1

PollEv.com/kaimengtan612

Let a and b be real constants, and let $U = \{(x, y) \mid ax^2 + by^2 = 0\}$.
Find conditions on a and b for U to be a subspace of $\mathbb{R}^2$.

Question 1

PollEv.com/kaimengtan612

Let a and b be real constants, and let $U = \{(x, y) \mid ax^2 + by^2 = 0\}$. Find conditions on a and b for U to be a subspace of $\mathbb{R}^2$.

Solution.

Case 1. $a = b = 0$: $U = \mathbb{R}^2$ in this case, which is a subspace of $\mathbb{R}^2$.

Question 1

PollEv.com/kaimengtan612

Let a and b be real constants, and let $U = \{(x, y) \mid ax^2 + by^2 = 0\}$. Find conditions on a and b for U to be a subspace of $\mathbb{R}^2$.

Solution.

Case 1. $a = b = 0$: $U = \mathbb{R}^2$ in this case, which is a subspace of $\mathbb{R}^2$.

Case 2. $a = 0$, $b \neq 0$: $U = \{(x, 0) \mid x \in \mathbb{R}\} = \operatorname{span}\{(1, 0)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Question 1

PollEv.com/kaimengtan612

Let a and b be real constants, and let $U = \{(x, y) \mid ax^2 + by^2 = 0\}$. Find conditions on a and b for U to be a subspace of $\mathbb{R}^2$.

Solution.

Case 1. $a = b = 0$: $U = \mathbb{R}^2$ in this case, which is a subspace of $\mathbb{R}^2$.

Case 2. $a = 0$, $b \neq 0$: $U = \{(x, 0) \mid x \in \mathbb{R}\} = \text{span}\{(1, 0)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Case 3. $a \neq 0$, $b = 0$: $U = \{0, y) \mid y \in \mathbb{R}\} = \text{span}\{(0, 1)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Question 1

PollEv.com/kaimengtan612

Let a and b be real constants, and let $U = \{(x, y) \mid ax^2 + by^2 = 0\}$. Find conditions on a and b for U to be a subspace of $\mathbb{R}^2$.

Solution.

Case 1. $a = b = 0$: $U = \mathbb{R}^2$ in this case, which is a subspace of $\mathbb{R}^2$.

Case 2. $a = 0$, $b \neq 0$: $U = \{(x, 0) \mid x \in \mathbb{R}\} = \operatorname{span}\{(1, 0)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Case 3. $a \neq 0$, $b = 0$: $U = \{0, y) \mid y \in \mathbb{R}\} = \operatorname{span}\{(0, 1)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Case 4. Both $a > 0$, $b > 0$ or both $a < 0, b < 0$: $U = \{(0, 0)\}$ in this case, which is a subspace of $\mathbb{R}^2$.

Question 1

PollEv.com/kaimengtan612

Let a and b be real constants, and let $U = \{(x, y) \mid ax^2 + by^2 = 0\}$. Find conditions on a and b for U to be a subspace of $\mathbb{R}^2$.

Solution.

Case 1. $a = b = 0$: $U = \mathbb{R}^2$ in this case, which is a subspace of $\mathbb{R}^2$.

Case 2. $a = 0$, $b \neq 0$: $U = \{(x, 0) \mid x \in \mathbb{R}\} = \text{span}\{(1, 0)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Case 3. $a \neq 0$, $b = 0$: $U = \{0, y) \mid y \in \mathbb{R}\} = \text{span}\{(0, 1)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Case 4. Both $a > 0$, $b > 0$ or both $a < 0, b < 0$: $U = \{(0, 0)\}$ in this case, which is a subspace of $\mathbb{R}^2$.

Case 5. $a < 0 < b$ or $b < 0 < a$: In this case, $(\sqrt{|b|}, \pm\sqrt{|a|}) \in U$, but $(\sqrt{|b|}, \sqrt{|a|}) + (\sqrt{|b|}, -\sqrt{|a|}) = (2\sqrt{|b|}, 0) \notin U$, so that U is not closed under addition, and hence is not a subspace.

Question 1

PollEv.com/kaimengtan612

Let a and b be real constants, and let $U = \{(x, y) \mid ax^2 + by^2 = 0\}$. Find conditions on a and b for U to be a subspace of $\mathbb{R}^2$.

Solution.

Case 1. $a = b = 0$: $U = \mathbb{R}^2$ in this case, which is a subspace of $\mathbb{R}^2$.

Case 2. $a = 0$, $b \neq 0$: $U = \{(x, 0) \mid x \in \mathbb{R}\} = \operatorname{span}\{(1, 0)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Case 3. $a \neq 0$, $b = 0$: $U = \{0, y) \mid y \in \mathbb{R}\} = \operatorname{span}\{(0, 1)\}$ in this case, and is hence a subspace of $\mathbb{R}^2$.

Case 4. Both $a > 0$, $b > 0$ or both $a < 0, b < 0$: $U = \{(0, 0)\}$ in this case, which is a subspace of $\mathbb{R}^2$.

Case 5. $a < 0 < b$ or $b < 0 < a$: In this case, $(\sqrt{|b|}, \pm\sqrt{|a|}) \in U$, but $(\sqrt{|b|}, \sqrt{|a|}) + (\sqrt{|b|}, -\sqrt{|a|}) = (2\sqrt{|b|}, 0) \notin U$, so that U is not closed under addition, and hence is not a subspace.

Conclusion: U is a subspace of $\mathbb{R}^2$ if and only if $ab \geq 0$. □

Question 2

PollEv.com/kaimengtan612

Let $V = \text{span}\{(1,1,0,0),(0,1,2,0),(0,2,0,1),(1,0,2,-1)\}$, and let $\boldsymbol{v} = (c,2,d,0)$. Find conditions on c and d such that $\{\boldsymbol{v},(0,2,0,1),(1,0,2,-1)\}$ is a basis for V.

Question 2

PollEv.com/kaimengtan612

Let $V = \text{span}\{(1,1,0,0),(0,1,2,0),(0,2,0,1),(1,0,2,-1)\}$, and let $\boldsymbol{v} = (c,2,d,0)$. Find conditions on c and d such that $\{\boldsymbol{v},(0,2,0,1),(1,0,2,-1)\}$ is a basis for V.

Solution.

$$\begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 0 & 2 & d \\ -1 & 1 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_4+R_1]{R_3-2R_1} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 1 & 1 & 0 & c \end{pmatrix} \xrightarrow{R_4-\frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} & c-1 \end{pmatrix}$$

$$\xrightarrow{R_4+\frac{1}{4}R_3} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & 0 & 0 & \frac{1}{2}c+\frac{1}{4}d-1 \end{pmatrix}$$

Question 2

PollEv.com/kaimengtan612

Let $V = \text{span}\{(1,1,0,0),(0,1,2,0),(0,2,0,1),(1,0,2,-1)\}$, and let $\boldsymbol{v} = (c,2,d,0)$. Find conditions on c and d such that $\{\boldsymbol{v},(0,2,0,1),(1,0,2,-1)\}$ is a basis for V.

Solution.

$$\begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 0 & 2 & d \\ -1 & 1 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_4+R_1]{R_3-2R_1} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 1 & 1 & 0 & c \end{pmatrix} \xrightarrow{R_4-\frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} & c-1 \end{pmatrix}$$

$$\xrightarrow{R_4+\frac{1}{4}R_3} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & 0 & 0 & \frac{1}{2}c+\frac{1}{4}d-1 \end{pmatrix}$$

Thus $\dim(V) = 3$, and $(c,2,d,0) \in V$ if and only if $\frac{1}{2}c + \frac{1}{4}d - 1 = 0$, i.e. $2c + d = 4$.

Question 2

PollEv.com/kaimengtan612

Let $V = \text{span}\{(1,1,0,0),(0,1,2,0),(0,2,0,1),(1,0,2,-1)\}$, and let $\boldsymbol{v} = (c,2,d,0)$. Find conditions on c and d such that $\{\boldsymbol{v},(0,2,0,1),(1,0,2,-1)\}$ is a basis for V.

Solution.

$$\begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 0 & 2 & d \\ -1 & 1 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_4+R_1]{R_3-2R_1} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 1 & 1 & 0 & c \end{pmatrix} \xrightarrow{R_4-\frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} & c-1 \end{pmatrix}$$

$$\xrightarrow{R_4+\frac{1}{4}R_3} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & 0 & 0 & \frac{1}{2}c+\frac{1}{4}d-1 \end{pmatrix}$$

Thus $\dim(V) = 3$, and $(c,2,d,0) \in V$ if and only if $\frac{1}{2}c + \frac{1}{4}d - 1 = 0$, i.e. $2c + d = 4$.

Also, when $(c,2,d,0) \in V$, then $\{(c,2,d,0),(0,2,0,1),(1,0,2,-1)\}$ is linearly independent if and only if $d \neq 2c$.

Question 2

PollEv.com/kaimengtan612

Let $V = \text{span}\{(1,1,0,0),(0,1,2,0),(0,2,0,1),(1,0,2,-1)\}$, and let $\boldsymbol{v} = (c,2,d,0)$. Find conditions on c and d such that $\{\boldsymbol{v},(0,2,0,1),(1,0,2,-1)\}$ is a basis for V.

Solution.

$$\begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 0 & 2 & d \\ -1 & 1 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_4+R_1]{R_3-2R_1} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 1 & 1 & 0 & c \end{pmatrix} \xrightarrow{R_4-\frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} & c-1 \end{pmatrix}$$

$$\xrightarrow{R_4+\frac{1}{4}R_3} \begin{pmatrix} 1 & 0 & 1 & 0 & c \\ 0 & 2 & 1 & 1 & 2 \\ 0 & 0 & -2 & 2 & d-2c \\ 0 & 0 & 0 & 0 & \frac{1}{2}c+\frac{1}{4}d-1 \end{pmatrix}$$

Thus $\dim(V) = 3$, and $(c,2,d,0) \in V$ if and only if $\frac{1}{2}c + \frac{1}{4}d - 1 = 0$, i.e. $2c + d = 4$.

Also, when $(c,2,d,0) \in V$, then $\{(c,2,d,0),(0,2,0,1),(1,0,2,-1)\}$ is linearly independent if and only if $d \neq 2c$.

Conclusion: $\{(c,2,d,0),(0,2,0,1),(1,0,2,-1)\}$ is a basis for V if and only if $2c + d = 4$ and $d \neq 2c$. □

Question 3

PollEv.com/kaimengtan612

Let $S = \{(1,1,0,0),(0,1,2,0),(0,2,0,1)\}$ and let $V = \text{span}(S)$.
Let $T = \{(1,2,2,0),(1,0,2,-1),(a,2,4-2a,0)\}$.
Find conditions on a such that T is a basis for V, and compute the transition matrix from S to T when T is a basis for V.

Question 3

PollEv.com/kaimengtan612

Let $S = \{(1,1,0,0),(0,1,2,0),(0,2,0,1)\}$ and let $V = \text{span}(S)$.
Let $T = \{(1,2,2,0),(1,0,2,-1),(a,2,4-2a,0)\}$.
Find conditions on a such that T is a basis for V, and compute the transition matrix from S to T when T is a basis for V.

Solution.

When T is a basis for V, then the transition matrix from $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ to T is $\begin{pmatrix}[\boldsymbol{v}_1]_T & [\boldsymbol{v}_2]_T & [\boldsymbol{v}_3]_T\end{pmatrix}$. To find $[\boldsymbol{v}_i]_T$, we use

$$\left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 2 & 2 & 4-2a & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3-2R1]{R_2-2R_1} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_4-\frac{1}{2}R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & a-1 & \frac{1}{2} & -\frac{1}{2} & 0 \end{array}\right) \xrightarrow{R_4+\frac{1}{4}R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Question 3

PollEv.com/kaimengtan612

Let $S = \{(1,1,0,0),(0,1,2,0),(0,2,0,1)\}$ and let $V = \text{span}(S)$.
Let $T = \{(1,2,2,0),(1,0,2,-1),(a,2,4-2a,0)\}$.
Find conditions on a such that T is a basis for V, and compute the transition matrix from S to T when T is a basis for V.

Solution.

When T is a basis for V, then the transition matrix from $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ to T is $\begin{pmatrix}[\boldsymbol{v}_1]_T & [\boldsymbol{v}_2]_T & [\boldsymbol{v}_3]_T\end{pmatrix}$. To find $[\boldsymbol{v}_i]_T$, we use

$$\left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 2 & 2 & 4-2a & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3-2R1]{R_2-2R_1} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_4-\frac{1}{2}R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & a-1 & \frac{1}{2} & -\frac{1}{2} & 0 \end{array}\right) \xrightarrow{R_4+\frac{1}{4}R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

From the three right columns of the final matrix, we see that S is linear independent, so that $\dim(V) = 3$.

Question 3: Solution (Cont'd)

$$\left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 2 & 2 & 4-2a & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Suppose $4 - 4a \neq 0$, i.e., $a \neq 1$. Then T is linearly independent and $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3 \in \operatorname{span}(T)$. Hence, $V = \operatorname{span}(S) \subseteq \operatorname{span}(T)$. Since $\dim(\operatorname{span}(T)) = |T| = 3 = \dim(V)$, we conclude that $\operatorname{span}(T) = V$ and that T is a basis for V. We can continue to get

$$\left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow[\frac{1}{4-4a}R_3]{-\frac{1}{2}R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & 1 & a-1 & \frac{1}{2} & -\frac{1}{2} & -1 \\ 0 & 0 & 1 & \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow[R_2-(a-1)R_3]{R_1-aR_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & \frac{a-2}{2(a-1)} & \frac{a}{2(a-1)} & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1-R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{a-2}{2(a-1)} & \frac{a}{2(a-1)} & 1 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Thus the transition matrix from S to T is $\begin{pmatrix} \frac{a-2}{2(a-1)} & \frac{a}{2(a-1)} & 1 \\ 0 & 0 & -1 \\ \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \end{pmatrix}$.

Question 3: Solution (Cont'd)

$$\left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 2 & 2 & 4-2a & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Suppose $4-4a \neq 0$, i.e., $a \neq 1$. Then T is linearly independent and $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3 \in \operatorname{span}(T)$. Hence, $V = \operatorname{span}(S) \subseteq \operatorname{span}(T)$. Since $\dim(\operatorname{span}(T)) = |T| = 3 = \dim(V)$, we conclude that $\operatorname{span}(T) = V$ and that T is a basis for V. We can continue to get

$$\left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow[\frac{1}{4-4a}R_3]{-\frac{1}{2}R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & 1 & a-1 & \frac{1}{2} & -\frac{1}{2} & -1 \\ 0 & 0 & 1 & \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow[R_2-(a-1)R_3]{R_1-aR_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & \frac{a-2}{2(a-1)} & \frac{a}{2(a-1)} & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1-R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{a-2}{2(a-1)} & \frac{a}{2(a-1)} & 1 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

Thus the transition matrix from S to T is $\begin{pmatrix} \frac{a-2}{2(a-1)} & \frac{a}{2(a-1)} & 1 \\ 0 & 0 & -1 \\ \frac{1}{2(a-1)} & -\frac{1}{2(a-1)} & 0 \end{pmatrix}$.

Question 3: Solution (Cont'd)

$$\left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 2 & 2 & 4-2a & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|ccc} 1 & 1 & a & 1 & 0 & 0 \\ 0 & -2 & 2-2a & -1 & 1 & 2 \\ 0 & 0 & 4-4a & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

If $4 - 4a = 0$, i.e., $a = 1$, then the matrix is reduced to

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -2 & 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $\boldsymbol{v}_1, \boldsymbol{v}_2 \notin \operatorname{span}(T)$, and so T cannot be a basis for V.

Next Lesson ...

We will revise **Chapters 5–6**.